Skills Summary

Writing and Using Piecewise Rules for Rates and Fees

Use this reference while completing your worksheet or when studying.

A fee schedule changes its rate at a **breakpoint**, so one linear rule cannot describe it. A piecewise rule gives one formula per region, each with the interval of inputs it governs. Every branch that starts charging only past a breakpoint has the same shape: **(what you already owe at the breakpoint) + (rate)(input − breakpoint)**. Subtracting the breakpoint is the whole trick — it counts only the units that are actually billed at the new rate.

Skill 1 — Writing a fee rule in pieces

Rule: flat fee F up to breakpoint B, then rate R per unit after:

$$C(x) = \begin{cases} F, & 0 < x \leq B \\ F + R(x - B), & x > B \end{cases}$$

Name the input and its unit first. Then decide which interval the question's input falls in *before* computing.

Example: A bike share charges \$2.00 to unlock, which covers the first 10 minutes; every minute after that costs \$0.15. What does a 26-minute ride cost?
Step 1. The ride is longer than the 10 included minutes, so it lands on the second branch — charge the unlock fee plus the minutes past 10.
Step 2. $C(m) = 2 + 0.15(m - 10)$ for $m > 10$, and $C(26) = 2 + 0.15(26 - 10) = 2 + 0.15 \cdot 16 = 2 + 2.40$.
$\boldsymbol{C(26) = 4.40}$ dollars

Try it: A scooter share charges \$3.00 to unlock, covering the first 5 minutes, then \$0.20 per minute after. Find the cost of a 20-minute ride in dollars.

check: $\boldsymbol{C(20) = 6.00}$

Skill 2 — Working a cost back to its input

Rule: to go from a total back to the input, first decide which branch the total belongs to (compare it with the flat fee), then set *that* branch equal to the total and solve. From $F + R(x - B) = T$ you get $x = B + \dfrac{T - F}{R}$ — undo the flat fee, undo the rate, then add the breakpoint back.

Example: A meter charges \$1.50 for the first hour and \$0.75 for each additional hour. A driver paid \$4.50. How many hours did they park?
Step 1. \$4.50 is more than the \$1.50 first-hour charge, so the payment came from the second branch — set that branch equal to 4.50 and solve for h.
Step 2. $1.50 + 0.75(h - 1) = 4.50 \Rightarrow 0.75(h - 1) = 3 \Rightarrow h - 1 = 4$.
$\boldsymbol{h = 5}$ hours

Try it: A tool library charges \$6.00 for the first 3 days and \$2.00 for each extra day. A member paid \$18.00. Find the number of days d.

check: $\boldsymbol{d = 9}$

Skill 3 — Finding a missing rate in a branch

Rule: when the per-unit rate is unknown, keep it as a letter and write the branch anyway: $F + r(x - B) = T$. One known input–output pair is enough, because the branch is linear in r: $r = \dfrac{T-F}{x-B}$. Read the answer as “dollars per unit” and say the unit out loud.

Example: A pottery studio charges \$15.00 for the first 2 hours and an unknown r dollars for each additional hour. A 6-hour session costs \$39.00. Find r.
Step 1. The 6-hour session is past the 2-hour breakpoint, so it is priced by the second branch — write that branch with r in it and solve the resulting one-variable equation.
Step 2. $15 + r(6-2) = 39 \Rightarrow 4r = 24$.
$\boldsymbol{r = 6.00}$ dollars per additional hour

Try it: A kayak rental costs \$20.00 for the first hour and r dollars for each extra hour. A 5-hour rental costs \$52.00. Find r in dollars per hour.

check: $\boldsymbol{r = 8.00}$

Skill 4 — Comparing two fee schedules

Rule: to find where two schedules charge the same, write one equation per schedule with a shared cost variable y, using each schedule’s branch for the region you care about, and solve the system. Then **check that the answer lies inside that region** — a solution outside it means the two schedules never meet there.

Example: Studio A charges \$18.00 for the first 3 sessions and \$2.00 per additional session. Studio B charges \$4.00 per session with no flat fee. For more than 3 sessions, when do they cost the same?
Step 1. Past 3 sessions each studio has one linear rule, so set up a system with a shared cost y and solve by substitution.
Step 2. $y = 18 + 2(n-3)$ and $y = 4n$, so $4n = 12 + 2n \Rightarrow 2n = 12$, giving $y = 4 \cdot 6 = 24$; and $6 > 3$, so the answer is in range.
$\boldsymbol{n = 6,\ y = 24.00}$

Try it: Gym A charges \$30.00 for the first 4 hours of court time and \$5.00 per additional hour; Gym B charges \$7.00 per hour. For more than 4 hours, find the hours t and shared cost y in dollars.

check: $\boldsymbol{t = 5,\ y = 35.00}$

(!) Watch out: The rate past a breakpoint applies to $x - B$ units, never to all x of them. Charging every unit at the higher rate is the single most common error on these problems, and it always overstates the bill. Also give every branch its interval: a rule with no intervals is not a piecewise rule, and there is no way to tell which formula to use.